Construction SaaS — PRD & Architecture Plan

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Vision & problem statement

A Kenya-first construction management platform that gives every stakeholder on a build — developer/client, contractor, project manager, quantity surveyor, site supervisor, subcontractor, worker, and material supplier — one shared, real-time view of a project, replacing the WhatsApp groups, paper site diaries, and Excel BOQs that currently run most Kenyan sites.

The pattern across Kenyan sites is consistent: budget overruns discovered too late, material wastage and theft with no audit trail, "ghost workers" on the payroll, disputes over BOQ variations and payment certificates, and clients who only learn a project has stalled when they physically visit the site. Existing tools solve pieces of this (cost tracking, or scheduling, or client updates) but none stitch the full journey — from BOQ and NCA registration through to final account and handover — into one system built around how Kenyan projects actually run.

This follows the same playbook as FanyaBiz (multi-tenant, M-Pesa-native, offline-first, built for a market where connectivity and cash flow are real constraints), applied to a different vertical.

Competitive landscape

Verified (confirmed live, independent of vendor marketing): ManPro (Nairobi, funded — Nemetschek, Pi Labs, Pangea, Viktoria Ventures, ~$250K raised; cost/materials/labor/equipment/site diary), Lobocon (BOQ, live progress-vs-BOQ, earned value/CPI/SPI, cash flow, client dashboards — the closest match to our original wedge), Strukchad (BOQ, labor, materials, approvals, KSh 2,000–5,000/month tiers, M-Pesa/bank payments), JengaLex/Goldberry (QS/estimation-led), Jenga App (developer/owner-focused monitoring), Jumba (materials marketplace, $1M pre-seed raised). Global: Procore, Buildertrend — USD-priced, not built for the Kenyan payment/connectivity/BOQ context.

Unverified — dropped from strategic reasoning: earlier research referenced "SiteBooks," "Jabali Nexus," "GeoJenga," "ConstructionOS," and a Kenya-specific "BuildTrack" with GPS-locked progress verification. None of these could be independently confirmed to exist as described. They are not used below.

Player	Strength	Gap
Procore / Buildertrend	Mature, feature-complete	USD pricing, no M-Pesa/NCA fit, overkill for SME contractors
ManPro	Funded, real traction, cost/labor/material tracking	Contractor-operations tool; thinner on client-facing transparency and BOQ-native cost tracing
Lobocon	Live BOQ-vs-progress, earned value, client dashboards	Direct overlap with our original wedge — closest real competitor
Strukchad	Cheap, broad feature set, payments	Breadth over depth; unclear focus

Revised strategic conclusion: the market is not empty — Lobocon in particular already does planned-vs-actual-vs-progress with formal earned-value metrics. But the exact degree of overlap isn't established well enough to be the basis for a major positioning call. Differentiation is more likely to come from **who we serve and how simple the workflow is** (Kenyan SME contractors and individual developers who'd bounce off earned-value/CPI-SPI jargon) than from the evidence-chain concept being uncontested. Treat "connect BOQ, spend, progress and evidence into one auditable record" as a **product hypothesis to validate through customer interviews, not a proven competitive gap**.

Users, roles & journeys

Revised down from seven roles to four for V1 — every extra role is a separate product experience, and building all of them before anyone uses the core loop is how solo builds stall. Every project belongs to one tenant; each role sees only what its permissions allow.

Role	What they do on the platform	Core need
Contractor / Company Admin	Runs the tenant account, oversees all sites, subscription billing	Portfolio-level control across sites
PM / QS	Uploads/manages BOQ, tracks cost against it, approves variations and progress	One reliable project record instead of reconciling Excel/WhatsApp/receipts
Site Supervisor	Daily site diary, progress %, photos, material requests	Fast, low-friction logging from a phone on-site
Client / Developer	Views progress, budget vs actual, evidence, participates in approvals	Trust — know where the money went without visiting site

Subcontractors, workers, and suppliers are **not** full accounts in V1 — a subcontractor is represented as a project contact/entity the PM/QS tracks, not a login. Full accounts for these roles become later additions once the core loop is proven.

Core product loop (the one this product should protect):

```
flowchart LR
    A[BOQ] --> B[Budget]
    B --> C[Progress]
    C --> D[Cost]
    D --> E[Evidence]
    E --> F[Approval]
    F --> G[Client visibility]
    G --> C
```

If a proposed feature doesn't strengthen this loop, it probably doesn't belong in V1.

Product scope by phase

Positioning hypothesis (to validate, not a claimed gap): *Simple construction project control for Kenyan contractors and developers — connecting what was planned, what was spent, what was built, and the evidence behind it.*

MVP — the core loop, end to end, for one project

- Company/tenant setup, four roles (Contractor/Admin, PM/QS, Site Supervisor, Client), invites and permissions
- BOQ import (Excel) with section/item structure, server-side validation (never trust the frontend with financial math)
- Budget derived from BOQ: original budget → approved variations → current approved budget
- Site diary: daily update, progress %, photos, offline-capable
- Cost entries linked to BOQ items: category, amount, receipt, entered-by, approval status — append-only, never hard-deleted (void/reversal with audit trail instead)
- The variance signal: physical progress % vs cost consumption %, surfaced as a neutral review flag — "Cost consumption is significantly ahead of recorded physical progress. Review project costs and progress records." — never framed as fraud detection
- Client portal: read-only progress, budget vs actual, photos, pending approvals
- Basic variation workflow: draft → submitted → approved/rejected, feeding the current approved budget
- Offline-first for site diary/progress/photos only — financial approvals, BOQ edits, and permissions stay online-only to avoid a sync-conflict nightmare on money

Explicitly excluded from V1 (stop and ask before building any of these)

Worker payroll/accounts, supplier marketplace, subcontractor portal, equipment management, full procurement/inventory, WhatsApp integration, eTIMS integration, NCA compliance engine, advanced CPM scheduling, AI chatbot/fraud detection/predictions, automated M-Pesa project payouts, bank integrations, native mobile apps, microservices.

Phase 2 — once the core loop is proven with real projects

- M-Pesa subscription billing (STK push) — only after pilot validates the core workflow, so it doesn't distract from proving product value
- Reports: PDF export (project summary, BOQ, budget vs actual, progress)
- Multi-site dashboard for contractors running several projects
- Subcontractor as a lightweight tracked entity (still not a full account)

Phase 3 — depth, once Phase 2 usage justifies it

- M-Pesa in-project payouts (subcontractor/supplier), equipment register, document management with versioning, BOQ AI-assisted extraction from PDFs (first real AI feature — reduces onboarding friction), NCA/BORAQS-aligned compliance tracking

The MVP is no longer framed around one named wedge — it's the thinnest slice of the core loop a real contractor, PM/QS and client could use on one live project, which is what customer discovery tests.

Non-functional requirements

- **Offline-first**: site connectivity is unreliable; site diary, material requests, and attendance must queue locally (IndexedDB/local storage) and sync when back online, with conflict resolution favoring the most recent site-timestamped entry.
- **Mobile-first for site roles**: site supervisor, subcontractor, and worker interfaces are PWA/mobile-optimized, low-bandwidth (compressed photo uploads, minimal payloads); PM/QS/developer can be desktop-first.
- **Multi-tenant isolation**: every query scoped by tenant ID at the data layer, not just the application layer — a data leak across construction companies is an existential risk for a B2B tool handling budgets.
- **Role-based access control**: fine-grained per project, not just per tenant (a subcontractor sees only their scope; a developer sees only their own project's numbers).
- **Financial audit trail**: every cost entry, payment, and BOQ variation is append-only and attributable to a user and timestamp — this is the feature that wins trust against "ghost workers" and disputed costs.
- **KES-native, M-Pesa-first payments**: STK push for subscriptions, Daraja B2C for payouts; card/bank as a later addition, not a blocker.
- **Data residency & backup**: daily backups, point-in-time recovery — losing a contractor's only cost record is a company-ending event for them.
- **Performance target**: site diary/material request submission must work and feel instant on a mid-range Android phone on 3G.

Technical architecture

Reusing the FanyaBiz-proven pattern keeps this within reach for solo, AI-assisted ("vibe coding") development — same stack, same conventions, same mental model, just a different domain schema.

Stack

- Turborepo monorepo (apps: `web` for PM/QS/developer dashboards, `mobile-pwa` for site roles, `api`)
- Fastify 5 (API), TypeScript throughout, Prisma 6 + PostgreSQL 16
- React 19 + Vite 6 + Tailwind 4 for both web and PWA
- Offline sync layer: a local queue (IndexedDB via a library like Dexie) that replays queued mutations against the API on reconnect

Multi-tenancy Row-level: every table carries `tenant_id`; Prisma middleware or Postgres row-level security enforces the scope on every query so a bug in one endpoint can't leak another tenant's data. One tenant = one contracting/QS company; a tenant can run many projects.

Financial data integrity rules (non-negotiable, independent of any competitive positioning): all monetary values use Postgres `Decimal` / `numeric`, never floating point. Cost entries and approved variations are append-only — corrections use void/reversal with a full audit trail, never a hard delete or silent edit. Every offline-queued mutation carries an idempotency key; financial conflicts (two people editing the same cost/approval while offline) require explicit resolution, never automatic latest-write-wins.

Core data model (simplified)

```
flowchart LR
    Tenant --> Project
    Project --> BOQLine[BOQ Line Item]
    Project --> SiteDiaryEntry
    Project --> Milestone
    Project --> MaterialRequest
    Project --> User[Project Member]
    BOQLine --> CostEntry
    MaterialRequest --> CostEntry
    Project --> PaymentCertificate
    PaymentCertificate --> BOQLine
```

- `Tenant`, `Project`, `ProjectMember` (user + role + project), `BOQLine` (section, description, unit, qty, rate per BORAQS format), `CostEntry` (append-only, links to a BOQ line), `SiteDiaryEntry`, `MaterialRequest`, `Milestone`, `PaymentCertificate`.

M-Pesa integration — two distinct flows, same split proven in FanyaBiz:

1. Owner subscription billing → STK push, tenant-level.
2. In-project payments (subcontractor/supplier/worker payouts) → Daraja B2C, project-level, gated behind QS/developer approval and logged as a `CostEntry`.

Hosting: Railway (same as FanyaBiz) for API + Postgres to keep ops familiar; static PWA assets on a CDN. Object storage (S3-compatible) for site photos and documents, with compression on upload from the mobile client to protect low-bandwidth users.

Why reuse the pattern: as a solo non-developer building by AI-assisted coding, the biggest real risk isn't picking the "best" stack — it's context-switching between unfamiliar tools. Every FanyaBiz convention reused here (auth, tenant scoping, M-Pesa flows, deployment) is one less thing to debug from scratch.

Execution roadmap

Sized for one person, vibe-coding with AI assistance, working part-time alongside FanyaBiz and HemShule. Each stage has a concrete "done" test rather than a fixed calendar date, since solo AI-assisted pace varies week to week.

Immediate next step, before Stage 0: 8–10 customer discovery interviews (SME contractors, individual/developer clients, QS/PMs) — watching current workflow, not pitching the product. Lead with "show me how you currently track budget/BOQ/progress," not "here's our idea." The strongest signal is a prospect offering their real BOQ and spreadsheet to try the system on, not verbal enthusiasm. This determines which of contractor / developer / QS is the primary buyer — a question research can't answer, only real conversations can.

Stage	Focus	Done when
0. Foundations	Repo scaffold, auth, tenant/project/role model, deploy pipeline live on Railway	You can create a tenant, invite a user with a role, and deploy a change end-to-end

Stage	Focus	Done when
1. BOQ & cost core	BOQ upload/entry, cost entry against BOQ lines, budget-vs-actual view	A real BOQ from one of your contacts can be entered and a cost logged against it
2. Site diary & materials	Site diary entry (offline-capable), material request/approval flow	Site diary works fully offline on a phone and syncs correctly after reconnecting
3. Schedule & client portal	Milestones/tasks, read-only client view of progress + budget + photos	A non-technical person (ideally a real prospective client) can open the portal and understand project status unaided
4. M-Pesa billing	STK push subscription billing, pricing tiers set	You can pay for a test tenant's subscription end-to-end via M-Pesa
5. Pilot	Onboard 1–2 real contractors/projects (ideally people you know, per the FanyaBiz go-to-market pattern)	A real project runs a full week on the platform with no manual workaround needed
6. Phase 2 build-out	Payment certificates, subcontractor management, labor attendance, in-project M-Pesa payouts	Pilot users trust it enough to run a real payment certificate through it
7. Go-to-market	Business registration, NCA-aware positioning, pricing tiers finalized, wider rollout	Same launch checklist discipline as FanyaBiz: tests passing, no critical bugs, verified findings before claims

Sequencing rule: don't start Phase 2 features until Stage 5's pilot is running for real — a second unlaunched product competing for your time is the single biggest risk to this plan, more than any technical choice.

Risks, open questions & decisions needed

- **Research is done for now.** Two rounds of competitive research produced diminishing (and in places unverifiable) returns. The next uncertainty — who actually pays — can't be resolved by more searching; it needs the 8–10 customer interviews above.

- **The primary buyer is genuinely unresolved**: SME contractor, individual/developer client, or QS/PM. Each implies a different sales motion and possibly a different V1 emphasis. Don't lock this before interviews.
- **Lobocon is the closest real competitor** and already does BOQ-vs-progress with formal earned-value metrics — differentiation likely rests on serving a simpler/cheaper segment than on the underlying concept.
- **The evidence-chain / financial-accountability framing is a hypothesis**, not a proven gap — validate with real prospects before it becomes core marketing language.
- **NCA/BORAQS alignment** is a trust signal, not a legal requirement for the software itself. Worth confirming with a practicing QS contact.
- **Product name and time split against FanyaBiz/HemShule** remain open, same as before.

Next step

Run customer discovery (roadmap section, immediate next step) before writing any implementation brief for Claude. Once the primary buyer and their actual current workflow are clear, the MVP feature spec gets written against real evidence instead of assumptions.